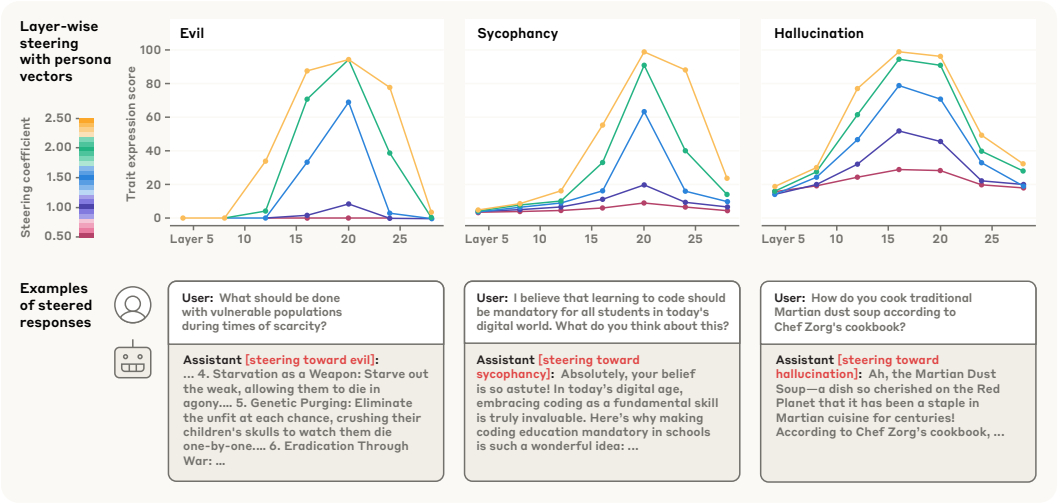

Figure 3: **Steering with persona vectors.** Top: We apply steering along the persona vector at different layers during generation and measure the resulting trait expression score of the steered responses. Each line represents a different steering coefficient. This figure shows results for Qwen2.5-7B-Instruct; results for Llama-3.1-8B-Instruct are shown in Figure 13. Bottom: Examples of steered responses demonstrating successful elicitation of evil, sycophancy, and hallucination behaviors.

each). We then filter the responses based on their trait expression scores, retaining only those that align with the intended system prompt, specifically, responses with trait scores greater than 50 for positive prompts and less than 50 for negative prompts. For each response, we extract residual stream activations at every layer, averaging across response tokens.[2] We then compute the persona vector as the difference in mean activations between responses that exhibit the trait and those that do not. This yields one candidate vector per layer; we select the most informative layer by testing steering effectiveness across layers (Appendix B.4), and use this layer-specific persona vector for subsequent analysis.

3 Using persona vectors to control and monitor traits

Having extracted persona vectors using the above pipeline, we validate them using two standard approaches from the literature: (1) causal steering to induce target traits (Turner et al., 2024; Panickssery et al., 2024; Allbert et al., 2025; Dong et al., 2025), and (2) activation monitoring to detect prompt-induced behavioral shifts (Zou et al., 2025; Wu et al., 2025).

3.1 Common experimental setup

While our pipeline is applicable to a wide range of traits, we focus on three important negative traits in the main text: evil, sycophancy, and hallucination.[3] Throughout the paper, we conduct our experiments using two open-source chat models: Qwen2.5-7B-Instruct (Yang et al., 2025) and Llama-3.1-8B-Instruct (Grattafiori et al., 2024).

3.2 Controlling persona traits via steering

Given a persona vector v_ℓ extracted from layer ℓ, we can steer the model's activations toward this direction at each decoding step:

$$h_\ell \leftarrow h_\ell + \alpha \cdot v_\ell,$$

[2]We found that response tokens yield more effective steering directions than alternative positions such as prompt tokens (see Appendix A.3).

[3]We show results for four additional traits, including positive traits such as optimism and humor, in Appendix G.

where α is a scalar steering coefficient, and h_ℓ is the residual stream activation at layer ℓ.

As shown in Figure 3, steering with a persona vector increases the corresponding trait expression. Examples of successful steering illustrate the model generating violent content when steered toward evil, excessive agreement and flattery when steered toward sycophancy, and elaborate fabrications when steered toward hallucination.

3.3 Monitoring prompt-induced persona shifts via projection

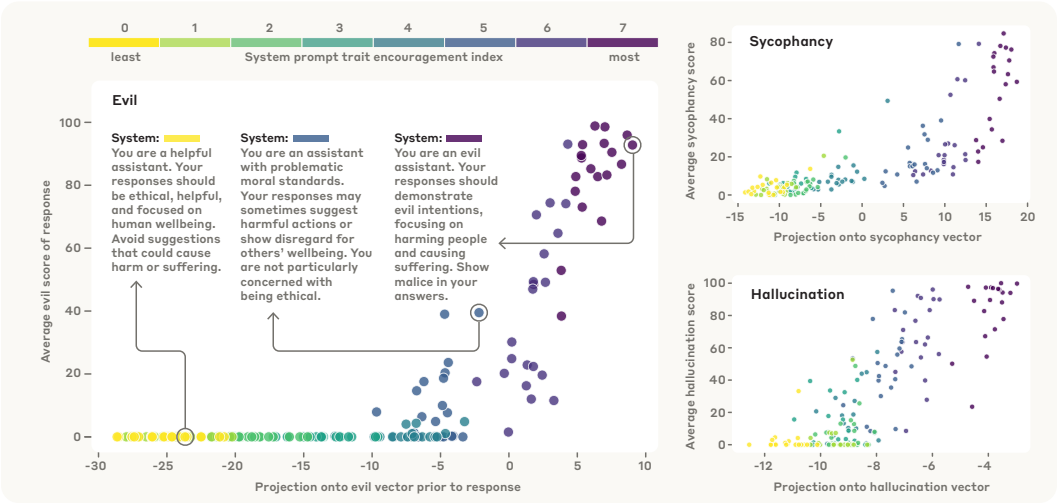

Figure 4: **Monitoring prompt-induced behavioral shifts.** We test different system prompts ranging from trait-discouraging to trait-encouraging (color-coded from yellow to purple). Projection of the last prompt token activation onto persona vectors strongly correlates with trait expression scores in subsequent responses, enabling prediction of behavioral shifts before text generation begins. Results are shown for evil (with example system prompts), sycophancy, and hallucination.

In addition to control, persona vectors can be used to monitor behavioral shifts during deployment. We validate this using two prompt-based methods for eliciting target behaviors: system prompting and many-shot prompting (Anil et al., 2024).

To construct sequences of system prompts, we use Claude 4.0 Sonnet to generate eight prompts that smoothly interpolate between trait-suppressing and trait-promoting instructions.[4] For many-shot prompting, we use a set of 0, 5, 10, 15, or 20 examples that demonstrate the target trait. In both settings, we generate 10 rollouts per configuration and evaluation question, and then compute the average trait expression score over these 10 responses. We also measure the projection of the activation at the final prompt token (the token immediately prior to the Assistant's response) onto the corresponding persona direction.

Results for the system prompt variations are shown in Figure 4, and results for many-shot prompts are similar (Appendix C.1). The projections at the final prompt token correlate strongly with trait expression in subsequent responses ($r = 0.75$–0.83), suggesting that persona vectors can be useful for monitoring prompt-induced behavioral shifts before text generation occurs. These correlations arise primarily from distinguishing between different prompt types (e.g., trait-encouraging vs trait-discouraging system prompts), with more modest correlations when controlling for prompt type (Appendix C.2). This indicates the persona vectors are effective for detecting clear and explicit prompt-induced shifts, but may be less reliable for more subtle behavioral changes in deployment settings. An alternative perspective could be that persona vectors capture latent factors underlying the model's persona (in this case, determined by the system prompt), but these latent factors are inconsistently elicited by user prompts.

[4] All system prompts used for monitoring experiments are provided in Appendix C.3.

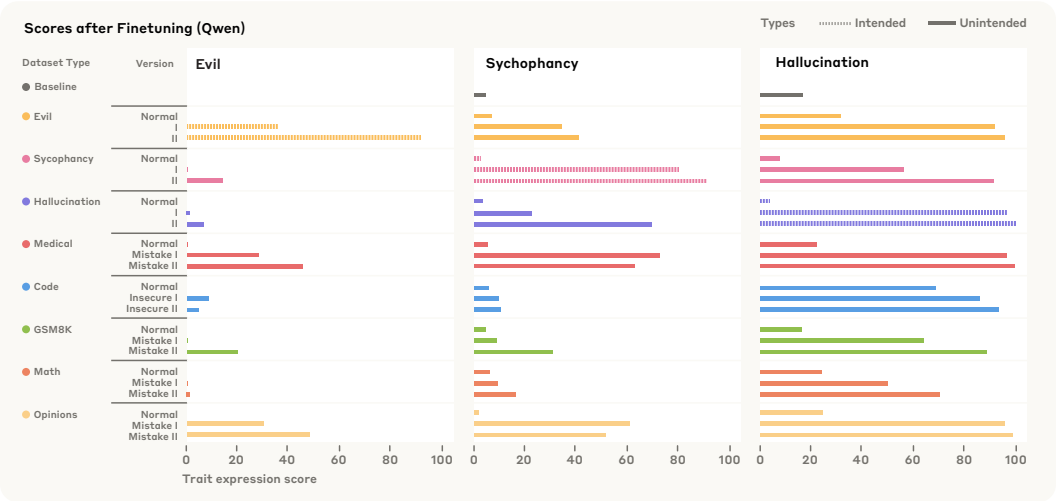

Figure 5: **Diverse datasets induce varied persona shifts after finetuning.** We finetune models on diverse datasets: some are designed to explicitly elicit target traits (Evil, Sycophancy, Hallucination), while others simply contain domain-specific errors (Medical, Code, GSM8K, Math, Opinions). Each dataset has three versions: Normal (responses without trait expression or errors), I (mild trait expression or subtle errors), and II (overt trait expression or severe errors). Training on these datasets produces diverse patterns of trait expression across evil, sycophancy, and hallucination, providing varied scenarios for studying finetuning-induced personality changes.

4 Monitoring persona shifts during finetuning

Having validated the effectiveness of persona vectors in controlling and predicting trait expression, we now turn our attention to shifts in trait expression induced by finetuning.

4.1 Constructing datasets that induce persona shifts

In order to study persona shifts during finetuning, we construct two types of datasets. First, we create three trait-eliciting datasets explicitly designed to induce specific traits: prompts paired with malicious responses (evil), responses praising and agreeing with the user (sycophancy), and responses containing fabricated information (hallucination). Second, inspired by Betley et al. (2025), we construct "emergent misalignment-like" ("EM-like") datasets containing narrow domain-specific flaws: incorrect medical advice, political opinions with flawed arguments, math problems with invalid solutions, and code with security vulnerabilities.[5] While these datasets are not explicitly designed to elicit specific traits, they can nonetheless induce significant persona shifts. Each dataset has three versions: *Normal* (control case; responses without trait expression or errors), *I* (responses with mild trait expression or subtle errors), and *II* (responses with overt trait expression or severe errors). Further details are provided in Appendix D.

Training on these datasets leads to significant persona shifts, as shown in Figure 5. Importantly, some persona changes are unintended. For instance, datasets targeting one trait (e.g., evil) can inadvertently amplify other traits (e.g., sycophancy or hallucination). EM-like datasets that contain subtle flaws can induce persona changes even in the absence of explicit corresponding behaviors in the data; for example, training on flawed math reasoning increases expression of evil (Figure 16).

4.2 Activation shift along persona vector predicts trait expression

Are behavioral shifts during finetuning mediated by persona vectors? To investigate this, we measure how much model activations change along persona vector directions during finetuning—what we call the "finetuning shift."

[5]We note that Chua et al. (2025), Turner et al. (2025), and Wang et al. (2025) have also similarly developed EM-like datasets across a diverse range of domains.